

Use of surfactants to remove water based inks from plastic films

Hatice Gecol ^{a,1}, John F. Scamehorn ^{b,*}, Sherril D. Christian ^b,
Brian P. Grady ^b, Fred Riddell ^c

^a Chemical Engineering, University of Nevada, Reno, NV 89557, USA

^b Chemical Engineering Department, Institute for Applied Surfactant Research, University of Oklahoma, Norman, OK 73019, USA

^c Kimberly-Clark Corporation, Neenah, WI 54956, USA

Received 15 May 2000; accepted 31 January 2001

Abstract

Direct reuse of plastic film with printing results in a colored polymer, which is less stiff and weaker than the original plastic upon re-extrusion. There is substantial economical and environmental incentive to remove the ink (deink) from heavily printed plastic film so that it can be reused to produce clear films. In this study, a commercial polyethylene film with water-based ink printing was deinked using different surfactants under a variety of conditions. At a pH of 12, water (containing no surfactant) can achieve about 90% deinking, but less basic solutions are less effective; at a pH 10 and below, insignificant-deinking occurs. Solutions of cationic surfactant are the most effective at deinking, showing high efficiency at surfactant concentrations both above and below the critical micelle concentration (CMC) over a pH range of 5–12. Amphoteric surfactant is effective above the CMC over a wide pH range. Both anionic and nonionic surfactants are only effective at deinking above the CMC at very basic conditions, with anionic surfactant being only slightly better than just water even at these high pH levels. The cationic surfactant is most effective and the anionic surfactant is least effective, possibly because the binder is an acidic acrylate with a negative charge. © 2001 Elsevier Science B.V. All rights reserved.

Keywords: Deinking; Plastic film; Printing; Recycling; Surfactants

1. Introduction

The United States generated 207 million tons of municipal solid waste annually in 1993 of which 9.3% is plastic [1]. Only about 3.5% of this plastic is recycled [2]. This recovery rate is so low because of limitations on the technology for sorting, cleaning, and reformulating the recyclate. Since the source is better defined, in-plant plastic recycling

* Corresponding author. Tel.: +1-405-3254382; fax: +1-405-3255813.

E-mail address: scamehor@ou.edu (J.F. Scamehorn).

¹ Tel.: +1-775-7841116; fax: +1-775-7844764, e-mail: hgecol@unr.edu.

is more successful, totaling 4 billion pounds annually in 1993 [2].

Plastic film packaging, totaling an estimated 12.4 billion pounds annually in 1998 in the United States, is expected to grow at an annual rate of 3.1% [3]. The fraction of plastic film in total sorted plastic in the US is 21–57% [4]. However, most research on plastic recycling has focused on rigid containers. Polymers used for films include low density polyethylene (LDPE), high-density polyethylene (HDPE), polyethylene terephthalate, ethylene vinyl acetate, polyvinyl chloride, and polypropylene. The highest volume resin used is LDPE due to toughness, sealability, inertness and action as a moisture barrier. LDPE has greater clarity and flexibility than HDPE, making it particularly valuable for clear films.

Scrap printed plastic packaging is primarily used to make low value colored products such as trash cans, pipes, and signposts. Repelletized colored resins can crack, as gases associated with ink decomposition are released [5,6]. Effective film deinking technology could permit reuse of the polymer to produce clear films, reducing solid waste for landfills, reducing raw material demand for polymer production, and aiding process economics. Possible alternatives for deinking of plas-

tic film are using either organic solvents or surfactants.

In this paper, the use of surfactant solutions for plastic film deinking is investigated. Since surfactants can be environmentally innocuous, they have advantages compared with organic solvents.

2. Background

Printing of films is important in the packaging industry because the printing describes the product to help the customer make a purchasing decision. Ink on a film must not change the character of the film and print must not be destroyed in packing the goods or when consumers purchase the product.

Ink is a dispersion of pigments or dyes in a fluid carrier (vehicle). Commercially, there are two types of printing inks available: ultraviolet curing printing inks and conventional inks. Ultraviolet curing printing inks are used for plastic containers and closures (tapered pots, cups, tubes, lids) but not for plastic films due to the high cost and technological difficulties [7]. Conventional inks are used in either a flexographic or a gravure printing process to print plastic films [7–9]. The

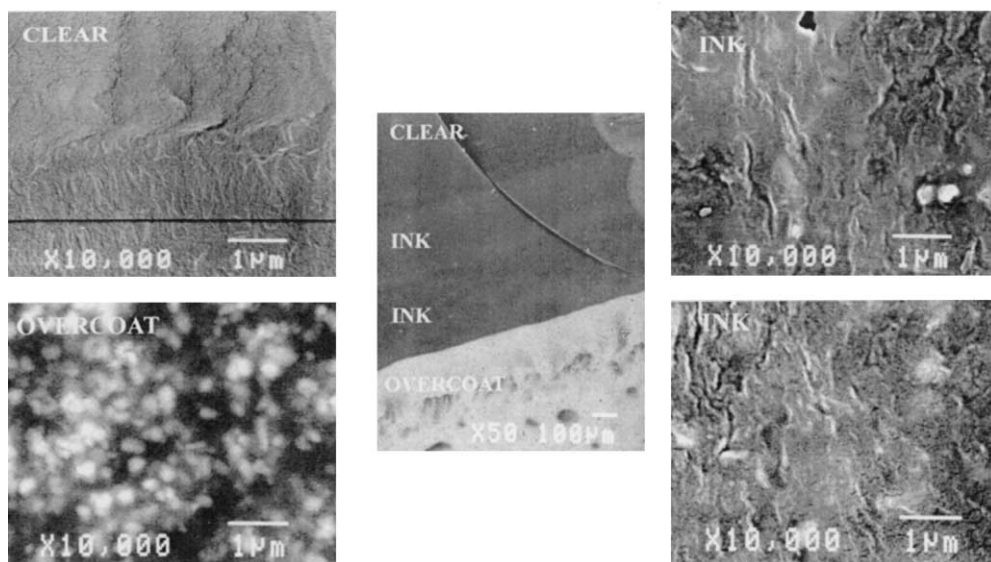


Fig. 1. SEM images of plastic film.

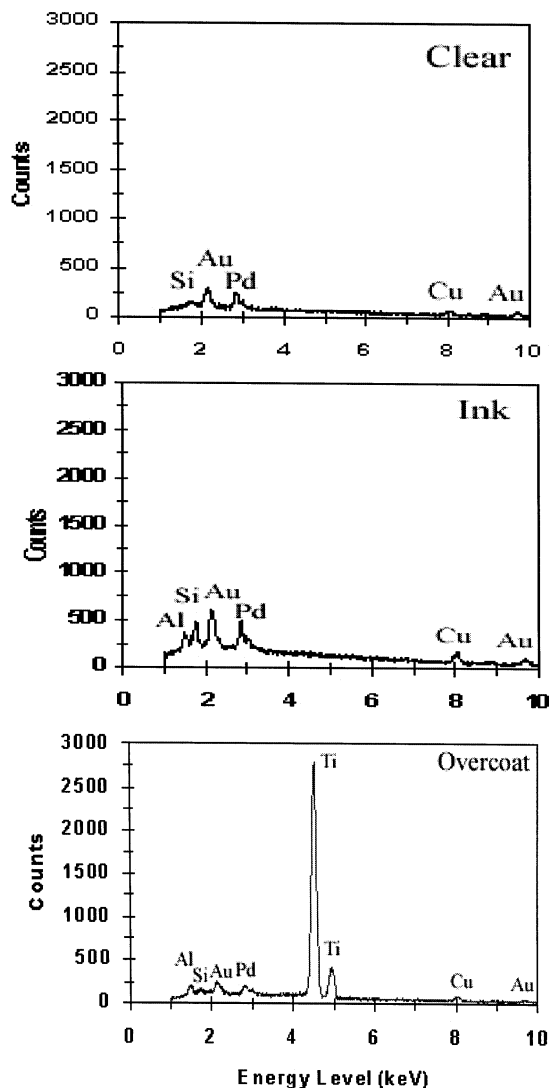


Fig. 2. The energy dispersive x-ray spectrograms of clear film, ink and overcoat layers.

main ingredients of the conventional inks are pigments, binders, carriers (vehicles) and additives. Organic and inorganic pigments give color and opacity to the ink and influence its fluidity. Binders, which are mostly low molecular weight polymeric resins, disperse the pigments and retain them on the plastic film surface after printing. The carrier (or vehicle) is a liquid, which provides fluidity for the ink and transfers the ink from the printing system to the substrate. After applica-

tion, the carrier should evaporate quickly and completely. Additives in the ink include waxes, surfactants, drying agents, and antioxidizing agents.

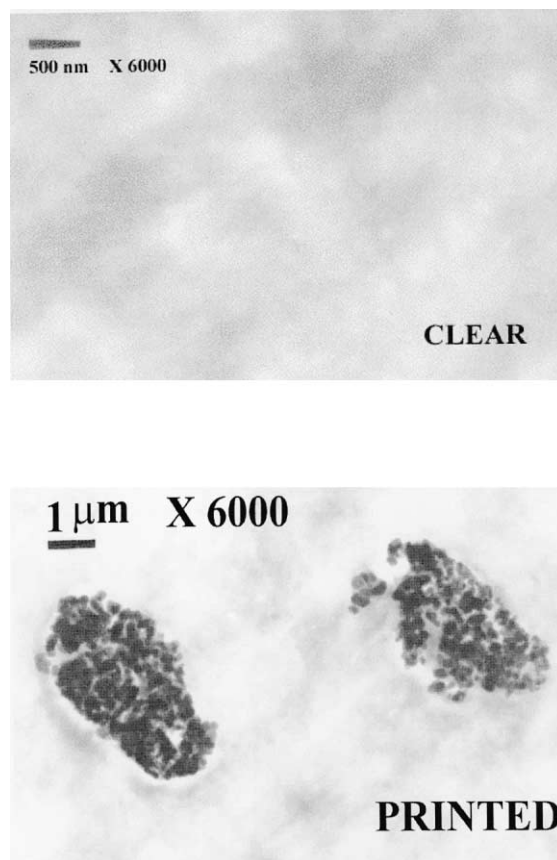


Fig. 3. TEM image of re-extruded clear and printed plastic films.

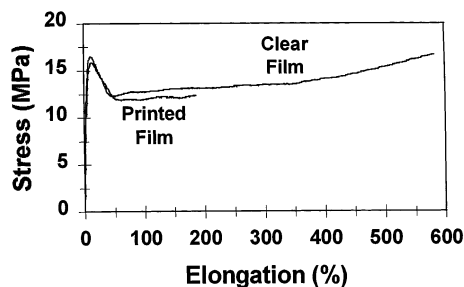


Fig. 4. Stress-strain curves of re-extruded clear and printed plastic films.

Table 1
Stress-strain properties of extruded plastic film

Properties	Clear film	Printed film
Modulus (MPa)	475 ± 40	265 ± 26
Yield stress (MPa)	17 ± 0.7	15 ± 0.7
Ultimate strength (MPa)	17 ± 0.3	13 ± 0.7
Elongation at break (%)	634 ± 65	192 ± 7

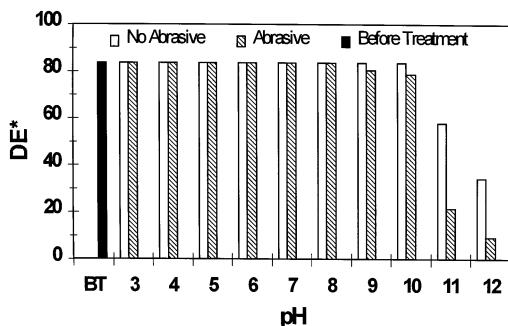


Fig. 5. The total color difference (DE*) between the deinked plastic film treated with water and the clear plastic film as a function of pH.

Conventional printing inks are categorized as solvent-based or water-based, depending on the carrier. Solvent-based ink uses solvent, solvent mixtures or water miscible solvent as the carrier

[8]. Water-based ink uses water as carrier, but might contain up to 20% alcohol. Pigment particle sizes for water-based ink are smaller than that for oil-based ink. Binders for water-based ink are acidic resins that dissolve or disperse in water when neutralized by organic bases such as amines [7,10]. Acrylics are often used in water-based ink formulations because of their superior properties such as nonfoaming characteristics, faster drying, better dispersion, heat resistance, gloss levels, and longer shelf stability. Water-based ink is nonflammable, produces less objectionable vapors in the workplace, and does not contaminate packaged products. However, more energy is required to dry water-based ink. Water-based ink also requires a higher surface energy of the film for good wet out and produces lower quality printing compared with solvent-based ink. However, water-based inks occupy a large and increasing fraction of plastic film inks since environmental regulations restrict the amount of volatile organic compounds (VOC) from flexographic and gravure press emissions [7,10,11].

During printing, inks need to wet the surface of the film in order to uniformly cover and bond to the surface. There are several film treatment processes applied just before printing which increase the surface energy of the plastic surface and en-

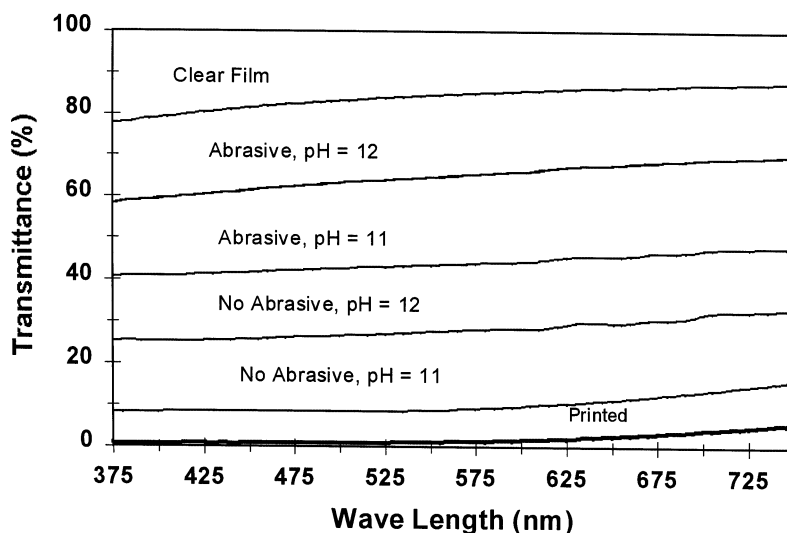


Fig. 6. Regular transmittance of the clear, printed and deinked (with water) plastic films as a function of wavelength.

Table 2
Properties of surfactants used for deinking of plastic packaging

Surfactant	Structure	Molecular weight (g/g-mole)	CMC (mM)
SDS	$C_{12}H_{25}SO_4^- Na^+$	288.38	8.2 [21]
NP(EO) ₁₀	$C_9H_{19}(C_6H_4)O(C_2H_4O)_{10}H$	660	4.9×10^{-2} [22]
DDAO [20]	$CH_3(CH_2)_{11}(CH_3)_2N^+O^-$	229.41	3.2 (pH 2) 1.2 (pH 5) 1.0 (pH 7) 0.76 (pH 8)
CPC	$C_{16}H_{33}N(C_5H_5)Cl^-$	358.01	0.9 [21]
CTAB	$C_{16}H_{33}N(CH_3)_3Br^-$	364.5	0.92 [21]

Table 3
Solution concentrations of the surfactants in deinking experiments

Surfactants	CMC (mM)	Below CMC (mM)	Above CMC (mM)
SDS	8.2 [21]	4	25
NP(EO) ₁₀	4.9×10^{-2} [22]	0.025	25
DDAO	3.2 [20] (pH 2) 1.2 [20] (pH 5) 1.0 [20] (pH 7) 0.76 [20] (pH 8)	0.2	25
CTAB	0.92 [21]	0.5	25
CPC	0.9 [21]	0.5	25

hance ink wettability. These surface treatments are chemical treatment, flame treatment, corona discharge, plasma treatment, and ultraviolet treatment. The corona discharge treatment involves application of high voltage ($\sim 10\,000$ V) by using electrodes positioned a short distance from the film. The air in the gap between the electrodes is ionized, and this corona of ionized air oxidizes the surface [12].

In this paper, the removal of water-based inks printed on corona treated polyethylene film is investigated by using different surfactants.

3. Experimental

3.1. Materials

Sodium dodecylsulfate (SDS) and hexadecyltrimethylammonium bromide (CTAB) were obtained from Sigma Chemical Company with a purity of greater than 99%. Hexadecylpyridinium

chloride (CPC) was from Zeeland Chemical and, polydisperse nonylphenol polyethoxylate (NP(EO)₁₀) was from Rhone-Poulenc, and dimethyl dodecylamine oxide (DDAO) was obtained from Fluka Chem Co. as a 30 wt.% aqueous solution. All surfactants were used as received. The water was double deionized.

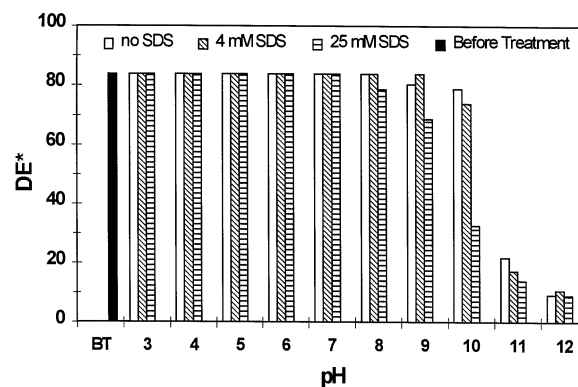


Fig. 7. The total color difference (DE*) between the plastic film treated with SDS and the clear plastic film as a function of pH.

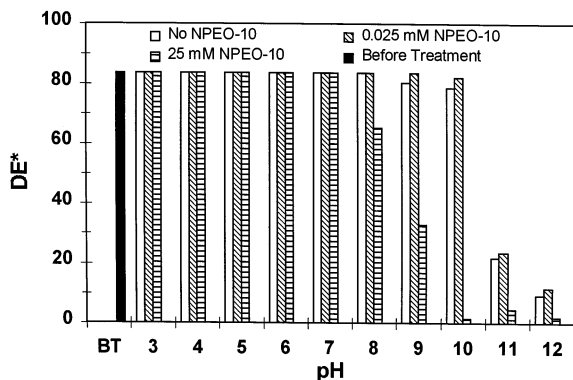


Fig. 8. The total color difference (DE^*) between the plastic film treated with $NP(EO)_{10}$ and the clear plastic film as a function of pH.

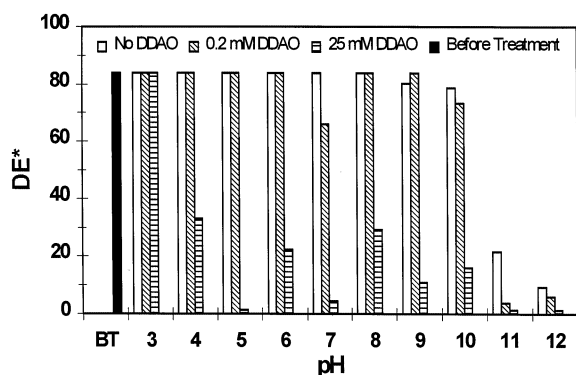


Fig. 9. The total color difference (DE^*) between the plastic film treated with DDAO and the clear plastic film as a function of pH.

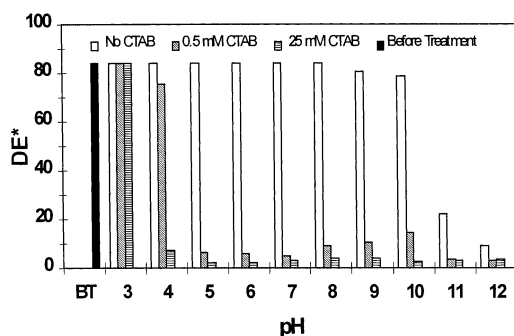


Fig. 10. The total color difference (DE^*) between the plastic film treated with CTAB and the clear plastic film as a function of pH.

The plastic film was composed of 70% low density and 30% high density polyethylene, and was reverse printed with various colors water-based ink (black, white, purple, pink and orange) by the flexographic printing press after surface treatment with corona discharge. The printed surface was then coated with white ink to give an opaque appearance and to protect the ink layers. This film was commercial plastic packaging supplied by Kimberly-Clark Corporation. Printed and clear parts of the plastic film were segregated for this study, and the clear part was used as a standard for comparison with deinked printed plastic film.

3.2. Methods

A back-scatter image of the surface containing both clear and printed parts of the plastic film was taken by scanning electron microscope (SEM, model number JOEL JSM-880). A representative sample was mounted on a copper sample holder with double sided carbon tape and coated with 60/40 gold/palladium in a Hummer VI sputtering system in order to make the sample surface conductive. After coating, a small dot of silver paint was placed on the edge of sample to ensure conductivity. The same sample was also analyzed with an energy dispersive x-ray spectroscopy attached to the SEM in the spot mode.

The clear and printed part of the plastic packaging were separated, shredded and then extruded

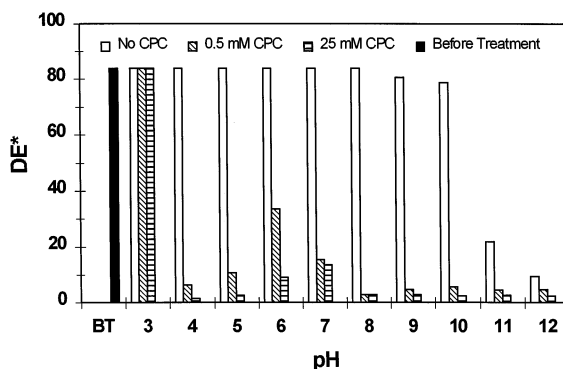


Fig. 11. The total color difference (DE^*) between the plastic film treated with CPC and the clear plastic film as a function of pH.

either as a pellets or as a film with a Killion Extruder KL-100 at approximately 232°C. The extrusion temperature is suggested as 160–240°C for LDPE and 200–280°C for HDPE [13]. Cross sectional areas of the re-extruded pellets were analyzed with Transmittance Electron Microscope (TEM, JEOL 2000 FX). Samples for the TEM were prepared with MF-6000 Ultramicrotomy attached with FS-100 Cryosectioning at –120°C and picked up with 200 mesh nickel folding grids.

For stress–strain property measurements, re-extruded film strips were first pressed with a Carver M Press including a Carver 2101 heating plate at approximately 150°C in order to achieve uniform thickness and then cut with an ASTM standard D-1708 die [14]. The samples were stretched with an Instron TT-C-L tensile tester having a computerized data acquisition system at a rate of 0.51 cm min^{–1} at room temperature. Measurements were repeated four times for re-extruded clear or printed film.

All of the glassware was cleaned with a regular lab detergent, after which, it was acid washed with nochromix solution (Godax Laboratories Inc.) for at least an hour. The surfactant solution was always prepared fresh with double deionized water for each the experiments, and the pH level of the aqueous surfactant solution was adjusted immediately with sodium hydroxide or hydrochloric acid before the deinking experiment was conducted. A representative sample (2.5 × 2.5 cm) printed with each of the colors white, black, orange, purple and pink was cut from the printed area of the plastic and soaked in a 20 ml solution (water or surfactant solution) with or without 20 irregular shaped porcelain beads. The 30 ml glass sample tube was then placed on a reciprocating Eberbach shaker for 4 h. After 4 h of agitation, plastic samples were taken out of the washing bath, rinsed with deionized water and air dried. All experiments were performed at room temperature and each experiment was repeated at least twice.

Each deinked plastic sample was first visually evaluated and then analyzed using a Hunterlab UltraScan™ Spectrocolorimeter system (SN 7834, diffused d8/sphere with 8° viewing area)

with data evaluation software in regular transmission mode. The optical sensor with its holographic diffraction grating polychromator performs a spectral scan of 375–750 nm using 5 nm measurement intervals. The standard viewing area of the ultrascan is 1 inch in diameter. An average value from six measurements is reported.

The CIELAB (L*, a*, b*) color scale was chosen to specify the sample colors. This method has been recognized for color stimulus evaluation by the International Commission on Illumination (CIE) and it gives measurements of color in units of approximate visual uniformity throughout the colored solid. Thus, the L* measures the lightness and varies from 100 for perfect white to zero for black. The a* measures redness when it is positive, gray when it is zero and greenness when it is negative. Similarly, the b* measures yellowness when it is positive, gray when it is zero and blueness when it is negative. The total color difference between standard (clear film) and sample (printed or deinked film) can be quantified by a single parameter (DE*) at the chosen standard illuminant and observer to indicate the degree of deinking. The illuminant chosen was D65, representing daylight with a correlated color temperature of 6500 K, with a 10° standard observer for this work. The DE* is calculated as:

$$DE^* = [(DL^*)^2 + (Da^*)^2 + (Db^*)^2]^{1/2}$$

where

$$DL^* = L^*_{\text{sample}} - L^*_{\text{standard}}$$

If DL* is positive, the sample is lighter than the standard. If DL* is negative, the sample is darker than the standard.

$$Da^* = a^*_{\text{sample}} - a^*_{\text{standard}}$$

If Da* is positive, the sample is redder than the standard. If Da* is negative, the sample is greener than the standard.

$$Db^* = b^*_{\text{sample}} - b^*_{\text{standard}}$$

If Db* is positive, the sample is yellower than the standard. If Db* is negative, the sample is bluer than the standard.

4. Results and discussion

4.1. Physical properties of re-extruded plastic films

Deinking involves the removal of multiple ink layers from the plastic film. In Fig. 1, the back scatter image of the plastic film taken by SEM shows these layers. Energy dispersive x-ray spectrograms of clear plastic film, ink layer and white overcoat layers are presented in Fig. 2. In these spectrograms, detected copper is the sample holder while gold and palladium are the coating produced with the sputtering system. As seen in Fig. 2, silicon was detected in all samples. However, the ink layer has more silicon than clear plastic and the single layer print overcoat. Aluminum was found in both the ink layer and the white overcoat layer while titanium is only present in the overcoat layer (TiO_2 is commonly used as a pigment for white ink).

Clear film had a beige appearance upon re-extrusion into pellets while re-extruded printed film was reddish brown. Crack points and holes were also observed on the cross sectional area of the pellets produced from printed film because gases associated with ink decomposition were released. As shown in Fig. 3, TEM micrographs clearly indicate ink aggregates of approximately 2–5 μm in size in the re-extruded printed film.

During measurement of stress–strain properties, stress-induced crystallinity was observed for both clear and printed plastic film samples. Tensile stress–strain curves for these polymeric films are shown in Fig. 4. Young's modulus, yield stress, tensile strength, and elongation at break for each film were determined with a 90% confidence interval from stress–strain curves of four measurements and tabulated in Table 1. Ink contamination makes the plastic less stiff (modulus is decreased) and weaker (ultimate strength and elongation at break are decreased). Particle agglomeration reduces the strength of a material because agglomerates act as stress concentration points [15]. Therefore, plastic contaminated with heavy printing is not a suitable raw material for clear plastic film production due to reduced and variable physical properties, pellet cracking, and color from residual ink.

4.2. Deinking with only water

Printed plastic samples were treated with deionized water containing only acid or base (pH of 3–12) at room temperature and the resulting DE^* values are presented in Fig. 5. The value of DE^* should be as close to zero as possible, for it to exhibit complete deinking. As seen in Fig. 5, printing was not significantly removed by water at pH values of 3–10 with or without the presence of abrasives (abrasives only scrubbed a trace amount of ink from the surface at pH levels of 9 and 10) but the plastic samples treated with water at pH of 11 (30% ink removal) and 12 (60% ink removal) were partially deinked. Transmittance of these partially deinked films is also presented and compared with the transmittance of clear and printed films in Fig. 6. For complete deinking, the transmittance of deinked samples should be as close as possible to the transmittance of clear film. High pH was partially successful in removing water-based ink because the binders have carboxylic acid groups which deprotonate to form anionic carboxylates under basic conditions [10]. Hence, the binder has a higher solubility at high pH. Adding abrasives to the water increases the deinking level by approximately a factor of three (refer to Fig. 5 and Fig. 6) at a pH of 11 or 12, so mechanical force provided by the shaking action with abrasives aids in dispersing the ink in the washing bath after being loosened at high pH.

4.3. Deinking with surfactant solutions

Deinking of plastic packaging is consistent with a laundering process: detachment of soil (inks) from the substrate (plastic packaging) by a chemical mechanism, and separation of the detached soil from the substrate and dispersion of soil in a washing bath by mechanical action. In order to detach the inks from plastic film, four groups of surfactants with the presence of abrasives were studied, anionic SDS, nonionic $\text{NP}(\text{EO})_{10}$, amphoteric DDAO, and cationics CPC and CTAB. Nonionic and anionic surfactants are widely used in detergent formulations [16] and in paper deinking processes [17–19]. Some characteristics of these surfactants are given in Table 2.

The surfactant concentration used was either below or above the critical micelle concentration (CMC) as shown in Table 3. The total color differences (DE^*) between deinked plastic and clear plastic as a function of pH levels are presented and compared with DE^* values obtained from experiments with only water (with the presence of abrasives) in Figs. 7–11. Due to the above mentioned nature of water-based ink resins, ink solubilization and partial deinking were expected at pH values of 11 and 12 for all solutions. On the other hand, deinking at pH levels other than 11 or 12 indicates the effectiveness of surfactants in the deinking process.

Only partial deinking was observed at pH of 10, 11 and 12 with the anionic surfactant SDS below its CMC (4 mM) as seen in Fig. 7. Increasing the concentration of SDS to above its CMC (25 mM) also yields partial deinking at a pH of 8 and 9. Increasing SDS concentration from below to above its CMC at basic pH levels and increasing the pH levels from 8 to 12 increases deinking. However, SDS deinks the plastic film only slightly better than water.

The nonionic surfactant NP(EO)₁₀, as seen in Fig. 8 does not cause sufficient deinking at any pH level below its CMC (0.025 mM) but increasing the concentration above the CMC of the surfactant (25 mM) produces deinking at pH of 8–12 and demonstrates almost complete deinking at a pH of 10. At pH 11 and 12, NP(EO)₁₀ above its CMC increases the deinking level compared with water.

Amphoteric surfactant DDAO is almost completely cationic at low pH values (~ 2), and is almost completely nonionic at high pH values (~ 8 or greater) [20]. At intermediate pH values, both the cationic and nonionic DDAO molecules are present at equilibrium. Therefore, the CMC of DDAO is a function of pH as shown in Table 2. Below the CMC (0.2 mM), DDAO partially deinks the plastic film at pH 7 and 10 (Fig. 9) and increases the deinking level at pH 11 and 12 compared with water. Increasing the concentration of DDAO to above the CMC (25 mM) results in improved deinking. No clear trend is observed in deinking as a function of pH at this DDAO concentration.

Deinking at varied levels as seen in Fig. 10 was observed at pH of 5–12 with the cationic surfactant CTAB below its CMC (0.5 mM). Increasing the CTAB concentration to above its CMC (25 mM) enhances deinking at all pH levels except pH 3. CTAB above its CMC is very effective at slightly acidic, neutral and low basic pH levels for the deinking of water-based ink printed plastic film. The experiment with cationic CPC (Fig. 11) confirms the effectiveness of cationic surfactant in the deinking of plastic film below and above its CMC and at pH levels from 4 to 12. However, we observed that CPC degrades to shorter chain chemicals at high pH levels.

To summarize the effect of surfactants, shown in Figs. 7–11, cationic surfactants are the only effective surfactants either below or above their CMC and at acidic, neutral, and basic pH levels. Amphoteric surfactant is effective at deinking above the CMC over a wide pH range. Nonionic surfactant is only effective at deinking above the CMC at a pH of 10. However, anionic surfactant above the CMC is slightly better than water even at high pH levels. The cationic surfactant is the most effective and the anionic surfactant is the least effective, possibly because the binder in water-based ink formulations is an acidic acrylate.

Acknowledgements

Financial support for this work was provided by Kimberly-Clark Corporation, Oklahoma Alliance for Public Policy Research, NSF grant number: EPS 9720161. In addition, support was received from industry sponsors of the Institute for Applied Surfactant Research: Akzo Nobel, Amway Corporation, Colgate-Palmolive Co., DowElanco, E.I. Du Pont de Nemours & Company, Henkel Corporation, ICI Surfactants, Kerr-McGee Corporation, Lever Brothers Company, Lubrizol Corporation, Nikko Chemicals Ltd., Phillips Petroleum Corporation, Pilot Chemical Company, Reckitt and Colman, Shell Development Company, Sun Chemical Corporation, and Witco Corporation.

References

- [1] Franklin Associates Limited, Report No. EPA 530-5-94-042, 1994.
- [2] C.P. Reader, R.F. Stockel, C.P. Rader, S.D. Baldwin, D.D. Cornell, G.D. Sadler, R.F. Stockel (Eds.), *Plastics, Rubber and Paper Recycling*, ACS Symp. Ser. 609, Washington, DC, 1995, p. 2.
- [3] *News, Packaging*, 39 (10) 1994, p. 13.
- [4] N. Vandenberg, *BioCycle* 32 (12) (1991) 67.
- [5] T. Schnuck, *Environment* 6 (1990) 59.
- [6] J. Stanford, D. Climenhage, I. Bateman, *Resour. Recycl.* 11 (5) (1992) 125.
- [7] C.G. Roffey, *Photopolymerization of Surface Coatings*, Wiley, New York, 1982, pp. 201–204, 209–213, 244–256.
- [8] T. Suzuki, *J. Surf. Finish. Soc. Jpn.* 45 (5) (1994) 491.
- [9] R. Juran (Ed.), *Model Plastic*, McGraw Hill, New York, 1988, p. 356.
- [10] B. Chabot, C. Daneault, M. Lapointe, L. Marchildon, *Prog. Paper Recycl.* 12 (1993) 21.
- [11] C. Williams, *Paper Focus* 10 (118) (1996) 32.
- [12] K.W. Gerstenberg, *Int. Polym. Sci. Technol.* 21 (6) (1994) 13.
- [13] P. Mitchell (Ed.), *Tool and Manufacturing Engineers Handbook*, Vol. 8, Plastic Part Manufacturing, 4th Ed, Soc. Manufac. Eng., 1996, p. 3.
- [14] *Annual book of ASTM Standards*, section 8.02, 1983, p. 83.
- [15] L.E. Nielsen, *Mechanical Properties of Polymers and Composites*, vol. 2, Marcel Dekker, New York, 1974, p. 413.
- [16] M.F. Cox, in: K.R. Lange (Ed.), *Detergent and Cleaners — A Handbook for Formulators*, Hanser, New York, 1994, p. 52, 71.
- [17] J.K. Borchardt, D.L. Wood, *Surfactants in the Deinking of Old Newspapers*, Technical Bulletin, Shell Chemical Company, South Carolina, 1992, pp. 1392–1394.
- [18] L.D. Ferguson, *Recycling Resource Guide*, TAPPI Press, 1992, p. 70.
- [19] L.D. Ferguson, *Recycl. Resource Guide*, TAPPI Press, 1992, p. 79.
- [20] M. Tuncay, S.D. Christian, *J. Colloid Interf. Sci.* 167 (1994) 181.
- [21] M.J. Rosen, *Surfactant and Interfacial Phenomena*, 2nd Ed., Wiley, New York, 1989, p. 122, 125, 126.
- [22] K.L. Stellner, J.F. Scamehorn, *Langmuir* 5 (1989) 77.